

Remote Asynchronous Phase Detection of Thoracoabdominal Breathing: A Novel Human Backscattering Model With FMCW Radar

Shuqin Dong¹, *Member, IEEE*, Wenren Zhou², *Graduate Student Member, IEEE*,
Xuelin Kong³, *Graduate Student Member, IEEE*, Shengjie Pan, Bo Wang⁴, *Member, IEEE*,
and Yongxin Guo⁵, *Fellow, IEEE*

Abstract—The detection of asynchronous thoracoabdominal movement is of great significance for the diagnosis of obstructive sleep apnea (OSA) and for determining the optimal time for weaning patients from mechanical ventilation in intensive care units. However, noncontact detection of this asynchronous phase requires radar systems with both high spatial resolution and high temporal fidelity to distinguish between the subtle, independent motions of the thorax and abdomen. This article proposes a novel detection approach based on frequency-modulated continuous wave (FMCW) radar utilizing a specific radar configuration. We first introduce an innovative human 4-D electromagnetic (EM) backscattering model that, grounded in physiological respiratory characteristics, accurately models the radar echoes generated by the distinct EM scattering from the thorax and abdomen. Building on this model, a robust FMCW baseband signal processing method is developed to enable the precise detection of the asynchronous phase of thoracoabdominal move-

ments. Simulations and experimental results demonstrate that the proposed model aligns well with real-world measurements, achieving a baseband signal waveform correlation exceeding 0.7, thus providing a solid theoretical foundation for related radar detection applications. Furthermore, clinical sleep studies involving 10 participants have validated that our proposed FMCW detection method successfully achieves noncontact detection of asynchronous thoracoabdominal movement, with the correlation between the estimated thoracoabdominal phase difference and the phase difference measured by the polysomnography (PSG) achieved a mean correlation of $r = 0.82$, showcasing its potential for medical application.

Index Terms—Asynchronous phase, 4-D electromagnetic (EM) backscattering model, frequency-modulated continuous wave (FMCW) radar, obstructive sleep apnea (OSA), thoracoabdominal movement.

Received 19 November 2025; revised 14 January 2026 and 9 February 2026; accepted 16 February 2026. This work was supported by the Startup Grant for Professor (SGP)—CityU SGP, City University of Hong Kong under Grant 9380170. (Shuqin Dong and Wenren Zhou contributed equally to this work.) (Corresponding authors: Bo Wang; Yongxin Guo.)

This work involved human subjects in its research. Approval of all ethical and experimental procedures and protocols was granted by the Human and Artefacts Ethics Sub-Committee of City University of Hong Kong under Application No. HU-STA-00001841 and the Medical Ethics Committee of The First Affiliated Hospital of Soochow University under Application No. 2024061.

Shuqin Dong is with the Department of Electrical Engineering, City University of Hong Kong, Hong Kong, China (e-mail: shuqdong@cityu.edu.hk).

Wenren Zhou is with the Department of Electrical and Computer Engineering, National University of Singapore, Singapore 119077, and also with the Center of Intelligent Sensing and Artificial Intelligence, National University of Singapore (Chongqing) Research Institute, Chongqing 401123, China.

Xuelin Kong is with the Department of Electrical and Computer Engineering, National University of Singapore, Singapore 119077, and also with the Research Platform for Biomedical and Healthcare Technology, National University of Singapore (Suzhou) Research Institute, Suzhou, Jiangsu 215123, China.

Shengjie Pan is with the Department of Neurology, The First Affiliated Hospital of Soochow University, Suzhou, Jiangsu 215006, China.

Bo Wang was with the Research Platform for Biomedical and Healthcare Technology, National University of Singapore (Suzhou) Research Institute, Suzhou, Jiangsu 215123, China. He is now with Sensori Healthcare Pte. Ltd., Singapore 118258 (e-mail: bowang@u.nus.edu).

Yongxin Guo is with the State Key Laboratory of Terahertz and Millimeter Waves and the Department of Electrical Engineering, City University of Hong Kong, Hong Kong, China (e-mail: yongxin.guo@cityu.edu.hk).

Digital Object Identifier 10.1109/TMTT.2026.3667509

I. INTRODUCTION

THE asynchronous phase of thoracoabdominal movement is a critical physiological indicator for diagnosing obstructive sleep apnea (OSA) and determining optimal weaning times for mechanical ventilation in intensive care units [1], [2]. While traditional contact-based sensors like respiratory inductance plethysmography (RIP) are clinically established [3], [4], they suffer from operational complexity, potential for displacement, and patient discomfort, making them ill-suited for long-term monitoring.

In recent years, radar-based sensing has emerged as a promising noncontact alternative [5], [6], [7]. Prior studies have demonstrated that continuous wave (CW) radar IQ trajectories can encode phase-offset information between thoracic and abdominal motion [6], [7], [8]. However, CW systems inherently lack range resolution [9]. Consequently, the received signal is a superposition of reflections from the thorax, abdomen, and surrounding clutter. While asynchronous motion can be inferred qualitatively from the distortion of the composite I/Q trajectory [7], CW radar cannot spatially resolve the thorax from the abdomen to extract their independent motion waveforms, making it difficult to directly obtain the phase difference parameters of thoracoabdominal motion. Conversely, while MIMO frequency-modulated CW (FMCW)

configurations can achieve spatial discrimination, they incur significant hardware complexity and cost [10], [11].

Recently, data-driven approaches incorporating AI technology have been employed to accurately identify OSA. Study [12] investigated the feasibility of using UWB radar for sleep apnea monitoring, extracting a single-channel respiratory signal and identifying OSA segments through amplitude reduction analysis. Study [13] proposed a method for detecting apnea-hypopnea events using a 60 GHz FMCW radar combined with a convolutional recurrent neural network. However, these studies lack a theoretical analysis of the feasibility of radar for thoracoabdominal respiration monitoring, and the introduction of AI further compounds this issue by reducing interpretability.

Furthermore, current electromagnetic (EM) modeling approaches—such as static ray-tracing or near-field analysis—fail to accurately simulate the dynamic, independent mechanics of thoracoabdominal respiration [14], [15], [16].

To overcome these limitations, this work first introduces an innovative human 4-EM backscattering model. In addition, we quantified the alignment between the proposed 4-D model and real-world measurement data, reporting a baseband signal correlation exceeding 0.7. Based on this theoretical basis, we propose a novel detection approach using a standard single-input single-output (SISO) FMCW radar. By employing a specific oblique-incidence geometry, we convert the spatial separation of the chest and abdomen into distinct range bins, enabling robust asynchronous phase detection without complex antenna arrays. The 10-patient clinical study demonstrates that the radar-derived thoracoabdominal phase difference maintains a mean correlation coefficient $r = 0.82$ with the polysomnography (PSG) gold standard.

The primary contributions of this work are summarized as follows.

- 1) *A Novel 4-D EM Backscattering Model*: We propose a dynamic human scattering model grounded in physiological respiratory mechanics. Unlike static models, this 4-D approach (3-D geometry + time) accurately simulates the independent EM scattering signatures of thoracic and abdominal movements, providing a theoretical foundation for complex respiratory waveform analysis.
- 2) *SISO Oblique-Incidence Strategy*: We demonstrate that a specific oblique radar configuration allows a simple SISO FMCW system to effectively resolve thoracoabdominal motion. This strategy leverages the intrinsic range resolution of FMCW to separate chest and abdomen signals into different range bins, balancing hardware simplicity with detection performance.
- 3) *Robust Asynchronous Phase Estimation Analysis*: We develop a robust baseband signal-processing framework and provide a comparative evaluation of Analytic, FFT-based, and correlation-based phase estimators. Simulations and clinical validation confirm that the proposed analytic method offers superior robustness for detecting the short-time asynchronous thoracoabdominal breathing events.

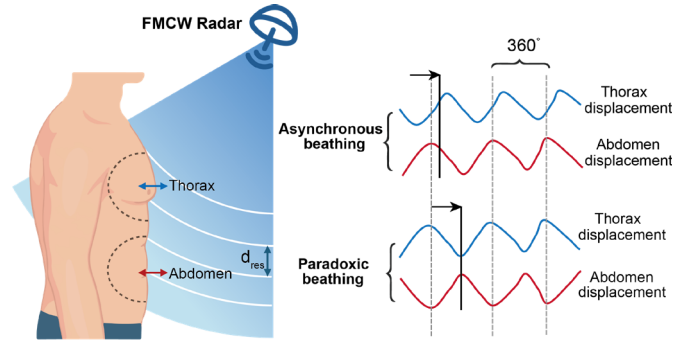


Fig. 1. Noncontact asynchronous phase detection of thoracoabdominal movement using FMCW radar with a distance resolution d_{res} .

II. HUMAN SCATTERING MODEL

A. Basis of Thoracoabdominal Respiratory Movement

In normal human respiration, thoracic and abdominal movements are coordinated synchronously. Efficient abdominal breathing, primarily driven by diaphragmatic contraction, facilitates relaxation and deep ventilation. In contrast, thoracic breathing relies more on intercostal muscle activity and is commonly observed during stressed states. Under conditions of increased respiratory load or muscle fatigue, asynchronous breathing may occur, characterized by a phase discrepancy between thoracic and abdominal movements—as illustrated on the right side of Fig. 1—which serves as an early indicator of declining respiratory efficiency. Further deterioration may lead to paradoxical breathing, where the thoracic cavity expands while the abdomen paradoxically retracts during inspiration [17].

Moreover, thoracoabdominal paradoxical respiration is a hallmark feature of OSA. This phenomenon arises from excessive inspiratory effort against upper airway obstruction, generating substantial negative intra-thoracic pressure and resulting in abnormal diaphragmatic motion. It represents a critical physiological marker for the diagnosis and severity assessment of OSA [18].

Therefore, independent observation of thoracic and abdominal respiratory movements, along with analysis of their waveform phase synchrony, constitutes an essential window for detecting and evaluating respiratory activity.

B. 4-D EM Backscattering Model of Human Chest

Based on the foregoing analysis, it is established that human respiration fundamentally comprises two distinct mechanical processes: thoracic breathing and abdominal breathing. Conventional EM scattering models employed to simulate respiration-induced modulation effects have predominantly treated the thorax as a unified, monolithic structure [5], an approach that constitutes an oversimplification inconsistent with physiological reality and consequently leads to inaccurate modeling.

In this work, we propose a novel physiologically grounded 4-D EM backscattering model that more faithfully represents the biomechanics of human respiration. The human thoracoabdominal region is geometrically modeled as a

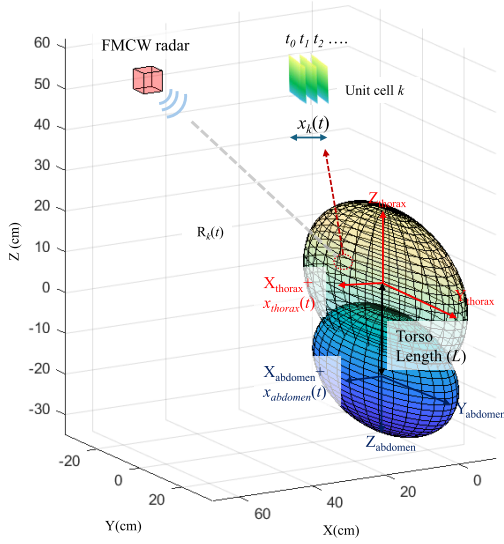


Fig. 2. 4-D EM backscattering model of human chest.

composite structure comprising two ellipsoids of disparate dimensions, as illustrated in Fig. 2. The larger ellipsoid represents the thoracic cavity, characterized by greater Y- and Z-axis radii, while the smaller ellipsoid represents the abdominal region. Based on anthropometric data from average adult populations, the semiaxis radii along the X-, Y-, and Z-directions for both ellipsoids are parameterized as X_{thorax} , Y_{thorax} , Z_{thorax} and X_{abdomen} , Y_{abdomen} , Z_{abdomen} for thoracic and abdominal compartments, respectively.

The 4-D human chest and abdomen scattering model proposed in this study aims to simulate radar echoes generated by human chest and abdomen respiratory movements. To balance the complexity of theoretical calculations with the fidelity of physical simulation, the model is based on the following core assumptions.

- 1) *Homogeneous Tissue*: At the operating frequency of 77 GHz, the EM wave penetration depth into human skin is extremely shallow. Consequently, the scattering is dominated by the surface boundary, making the assumption of a homogeneous dielectric (modeled via complex permittivity) sufficient for capturing phase modulations.
- 2) *Far-Field Condition*: In clinical settings like the ICU or sleep labs, the radar is typically positioned 0.65–1 m from the patient. At 77 GHz, this distance satisfies the far-field criteria, allowing the use of the physical optics (POs) method and plane-wave approximation for radar cross section (RCS) estimation.
- 3) *1-D Motion*: While human respiration is complex, the radial displacement toward the radar represents the primary component of the phase change. The model focuses on this longitudinal kinematic equation to ensure computational efficiency while maintaining high fidelity in asynchronous phase detection.
- 4) *No Multipath*: The model prioritizes the direct backscattering from the discretized unit cells. In a controlled hospital bed environment, the oblique geometry and the use of antennas with specific half-power beamwidths

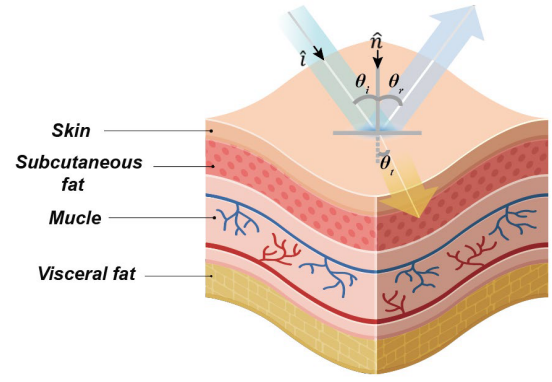


Fig. 3. Schematic of EM waves obliquely incident on the k th unit cell.

(HPBW)s minimize interference from static environmental reflections.

In the context of millimeter-wave radar detection, the surface of the composite ellipsoidal structure representing the thorax and abdomen is subdivided into numerous planar quadrilateral unit cells to approximate its geometry. This discretization facilitates the subsequent calculation of the RCS of this complex target. The respiratory kinematics are incorporated through empirically derived motion equations $x_{\text{thorax}}(t)$ and $x_{\text{abdomen}}(t)$, which define the time-varying displacements of thoracic and abdominal components [19], respectively,

$$x_{\text{thorax}}(t) = A_t \cos^{2n}(\pi f_R t), \quad n \in \mathbb{Z}^+ \quad (1)$$

$$x_{\text{abdomen}}(t) = A_a \cos^{2n}(\pi f_R t + \varphi), \quad n \in \mathbb{Z}^+ \quad (2)$$

where f_R is respiration frequency, n is a scalar parameter defining the waveform type (in the subsequent simulation, we choose n as 3 to generate the common respiratory plateau phase of the respiratory waveform), φ is phase difference of thoracic and abdominal movements, and A_t and A_a are the respiratory motion amplitudes of thorax and abdomen, respectively.

In order to further establish the EM wave backscattering model, it is necessary to calculate the RCS of the human chest model.

The PO method represents a high-frequency asymptotic approximation technique particularly suitable for electrically large objects. We employ the PO methodology to estimate the RCS of the human thoracoabdominal surface [20], [21]. For the k th unit cell, as illustrated in Fig. 3, an EM wave impinges on the discretized human skin surface at an incidence angle θ_i . Given the surface normal vector $\hat{n}$, the RCS of this specific skin patch is calculated as

$$\sigma_k = \frac{4\pi A^2}{\lambda^2} \cos^2(\theta_i) |R_s(\theta_i)|^2, \quad \text{where } \hat{n} \cdot \hat{i} > 0 \quad (3)$$

where A denotes the physical area of the unit cell, λ is the wavelength, and $\hat{i}$ is the unit vector of the incident wave direction, and contributions from the shadowed region are neglected according to the PO method. $R_s(\theta_i)$ is the Fresnel reflection coefficient, which supersedes the perfect electrical conductor (PEC) assumption to accurately model the dielectric properties of biological tissues.

It should be noted that, based on the radar used in the detection scenario and its configuration, the incident direction

vector $\hat{\mathbf{i}}$ is determined by the radar's mounting orientation. Furthermore, to enable more precise echo calculations, unit cells located outside the radar's HPBW are excluded from the analysis.

The experimental setup employs a 77 GHz radar with 4 GHz bandwidth, whose antenna exhibits a HPBW of 30° in the XZ plane of the coordinate system defined in Fig. 2, ensuring sufficient coverage of the thoracic and abdominal region. Furthermore, the target is positioned within the far-field region of the radar. At millimeter-wave frequencies, the energy penetration into deeper tissues beyond the skin layer is negligible; thus, this effect is justifiably disregarded in our model.

The EM properties of human tissue are modeled by its complex permittivity at the operational frequency of 77 GHz. Following established literature [22], the relative permittivity and effective conductivity of skin tissue are taken as $\epsilon_r = 8.5$ and $\sigma = 35$ S/m, respectively. The corresponding complex dielectric constant $\tilde{\epsilon}_2$ is then given by

$$\tilde{\epsilon}_2 = \epsilon_0 \epsilon_r - j \frac{\sigma}{\omega} \quad (4)$$

where ω is the angular frequency. Subsequently, the wave impedance $\tilde{\eta}_2$ within the biological tissue is derived as

$$\tilde{\eta}_2 = \sqrt{\frac{\mu_0}{\tilde{\epsilon}_2}}. \quad (5)$$

The reflection coefficient $R_s(\theta_i)$ is computed using the Fresnel equation. For the s-polarized component, pertinent to the presented model, it is expressed as

$$R_s(\theta_i) = \frac{\tilde{\eta}_2 \cos \theta_i - \eta_0 \cos \theta_t}{\tilde{\eta}_2 \cos \theta_i + \eta_0 \cos \theta_t} \quad (6)$$

where η_0 denotes the free space wave impedance, and θ_t denotes the complex angle of transmission, determined by applying the Snell's law in its complex form to account for the lossy dielectric nature of the skin, as shown in Fig. 3.

Based on the preceding analysis framework, the fundamental characteristics of FMCW radar detection necessitate consideration of range bins as discrete distance intervals rather than point targets. Each range bin encompasses multiple unit cells positioned at approximately equivalent radial distances from the radar transceiver. Considering the EM backscattering from unit cells located at radial distance R_k from the radar transceiver, the composite echo signal corresponding to the m th range bin after FMCW reception, down-conversion, and range-FFT processing can be mathematically formulated as follows:

$$X[m, t] = \sum_{k \in \mathcal{K}_m} A_k(t) \cdot \exp\left(-j \frac{4\pi R_k(t)}{\lambda}\right) \cdot W\left(m - \frac{2R_k B}{cT}\right) \cdot \exp(j\phi_k) + N[m] \quad (7)$$

where $\mathcal{K}_m$ is set of all unit cells whose radial distances R_k fall within the m th range bin interval, $W(\cdot)$ function is range-FFT spectral response due to sampling window, B is the bandwidth, and T the chirp duration of FMCW radar, ϕ_k is additional phase contributions from surface orientation and path variations, $N[m]$ is additive complex Gaussian noise in

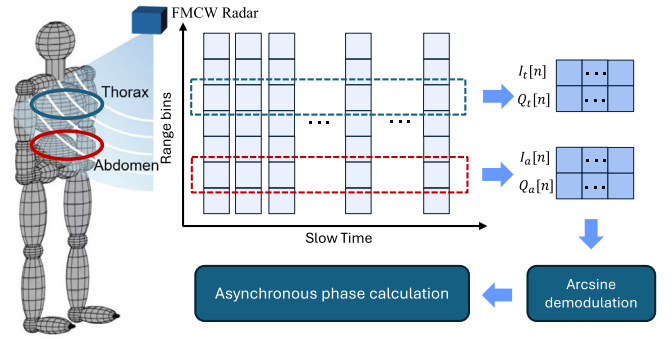


Fig. 4. Thoracoabdominal movement detection using FMCW radar.

the m th range bin, and A_k is amplitude term encompassing radar equation factors [23]

$$A_k(t) = \sqrt{\frac{P_t G_t G_r \lambda^2 \sigma_k(t)}{(4\pi)^3 R_k^4 L}} \quad (8)$$

where P_t is the radar transmit power, G_t is the transmit antenna gain, G_r is the receiving (RX) antenna gain, λ is the wavelength, and L is the free-space path loss.

Above formulation enables the generation of a dynamic 4-D (3 spatial dimensions + 1 temporal dimension) EM scattering model of respiratory mechanics. For each discrete time instance t_n , the model generates the complete geometrical configuration—position and orientation—of the entire discretized thoracoabdominal surface based on the predefined ellipsoidal dimensions and their associated kinematic equations. Subsequent integration with incident and reflected radar wave analysis yields a significantly more accurate radar return equation, facilitating enhanced fidelity in simulating physiological modulation effects in radar signatures.

C. Asynchronous Phase of Thoracoabdominal Movement Using FMCW Radar

This work will take advantage of the inherent range resolution and CW phase sensitivity of FMCW, and cleverly place the chest and abdomen under different range bins by obliquely incident on the surface of the thoracic cavity, as shown on the left side of Fig. 4.

For a classic 1 transmitting (TX)/1 RX FMCW radar, the transmitted signal within a single chirp interval is mathematically formulated as follows:

$$s_{Tx}(t) = \exp[j \cdot (2\pi f_c t + \pi \beta t^2 + \theta)] \quad (9)$$

where t denotes the “fast-time” variable within the chirp interval $[-T/2, T/2]$, f_c is the carrier frequency, $\beta = B/T$ is the chirp rate, θ is the initial phase.

The received signal is a summation of echoes from multiple targets. After de-ramping, the intermediate frequency (IF) beat signal for single target can be expressed as

$$s_b(t, \tau) = A \cdot \exp[j \cdot \{(4\pi \beta R(\tau)/c_0)t + (4\pi R(\tau)/\lambda) + \Delta\delta\}] \quad (10)$$

where A is a complex amplitude factor encompassing path loss and system gain, c_0 is the speed of light, τ is the “slow-time” index, representing the sequence of chirps, $R(\tau)$ is the

time-varying distance from the radar to the target, λ is the wavelength corresponding to f_c , $\Delta\delta$ is a residual phase term, often negligible in coherent radar systems.

A range-FFT is applied along the fast-time dimension t of the signal in (10), yielding the range profile

$$s_b(f, \tau) = A \cdot T \exp[j \cdot (4\pi R(\tau)/\lambda)] W(T(f - f_B)) \quad (11)$$

where $f_B = 2\beta R(\tau)/c_0$ is the beat frequency.

For multipoint target detection scenarios, such as independently monitoring chest and abdominal movements in vital sign monitoring, consider two distinct targets located at different ranges: the chest at range $R_{m1} = d_{\text{res}} \cdot m1$ and the abdomen at range $R_{m2} = d_{\text{res}} \cdot m2$, where $d_{\text{res}} = c_0/(2B)$ is inherent range resolution of FMCW. Given the range-FFT slow-time signal $s_b(f, \tau)$ obtained from (11), the complex signals extracted from these two specific rangebins can be mathematically expressed as follows:

$$\begin{aligned} S_{m1}(\tau) &= s_b(f_{m1}, \tau) \\ &= A_{m1} \cdot T \cdot \exp \left[j \cdot \frac{4\pi R_{m1}(\tau)}{\lambda} \right] \cdot W(T(f_{m1} - f_{B,1})) \end{aligned} \quad (12)$$

$$\begin{aligned} S_{m2}(\tau) &= s_b(f_{m2}, \tau) \\ &= A_{m2} \cdot T \cdot \exp \left[j \cdot \frac{4\pi R_{m2}(\tau)}{\lambda} \right] \cdot W(T(f_{m2} - f_{B,2})) \end{aligned} \quad (13)$$

where f_{m1} and f_{m2} are the beat frequencies corresponding to range bins $m1$ and $m2$, respectively, $f_{B,1} = 2\beta R_{m1}(\tau)/c_0$ and $f_{B,2} = 2\beta R_{m2}(\tau)/c_0$ are the instantaneous beat frequencies for each target, A_{m1} and A_{m2} are the respective amplitude factors for each target.

During this process, the amplitude of thoracic and abdominal respiratory motions should theoretically not exceed the range resolution d_{res} of the FMCW system. In practical scenarios, under oblique incidence conditions, the radial displacement component relative to the radar is significantly reduced. Typically, centimeter-level respiratory motions fall well within this range, ensuring effective phase demodulation without ambiguity.

Since vital sign motion represents a relatively slow displacement, meaning $f - f_B$ remains sufficiently small, the windowing function $W(T(f - f_B))$ can be approximated as unity. This yields the simplified signal

$$S_{m1}(\tau) = A_{m1} \cdot T \cdot \exp \left[j \cdot \frac{4\pi R_{m1}(\tau)}{\lambda} \right] \quad (14)$$

$$S_{m2}(\tau) = A_{m2} \cdot T \cdot \exp \left[j \cdot \frac{4\pi R_{m2}(\tau)}{\lambda} \right]. \quad (15)$$

Subsequently, equivalent CW I/Q signals can be obtained from these two signals, enabling the demodulation of both thoracic and abdominal respiratory motions using standard phase demodulation algorithms.

Following the successful demodulation of displacement signals $R_{m1}(\tau)$ and $R_{m2}(\tau)$ representing thoracic and abdominal motions, respectively, the phase relationship between these physiological signals can be quantified using the following phase estimation techniques.

1) *Proposed Analytic Method*: The normalized $R_{m1}(\tau)$ and $R_{m2}(\tau)$ are defined as

$$x_{\text{thorax}}(t) = \cos^2(\pi f_R t) \quad (16)$$

$$x_{\text{abdomen}}(t) = \cos^2(\pi f_R t + \varphi/2) \quad (17)$$

where $n = 1$, which is to simplify the calculation process, also aligns with the empirical observation that asynchronous breathing often has a shorter respiratory plateau [24]. φ is the phase difference between thoracic and abdominal respiratory movements.

Applying the power-reduction identity $\cos^2\theta = (1 + \cos 2\theta)/2$, the signals transform to

$$x_{\text{thorax}}(t) = \frac{1}{2} [1 + \cos(2\pi f_R t)] \quad (18)$$

$$x_{\text{abdomen}}(t) = \frac{1}{2} [1 + \cos(2\pi f_R t + \varphi)]. \quad (19)$$

The difference signal is obtained by subtraction

$$\begin{aligned} d(t) &= x_{\text{thorax}}(t) - x_{\text{abdomen}}(t) \\ &= \frac{1}{2} [\cos(2\pi f_R t) - \cos(2\pi f_R t + \varphi)]. \end{aligned} \quad (20)$$

Through mathematical derivation

$$d(t) = \sin(\varphi) \cdot \sin(2\pi f_R t + \varphi/2). \quad (21)$$

The resulting differential signal manifests as a sinusoidal waveform with frequency f_R and amplitude $A_d = |\sin(\varphi)|$. Consequently, the phase difference can be estimated through

$$\varphi_{\text{ana}} = 2 \cdot \arcsin(A_d) \quad (22)$$

where A_d represents the measured amplitude of the differential signal $d(t)$.

2) *Cross-Correlation Method*: The cross-correlation method can be described as follows: The cross-correlation function between the two displacement signals is computed as follows:

$$R_{xy}[k] = \frac{\sum_n (R_{m1}[n] - \mu_{R_{m1}})(R_{m2}[n+k] - \mu_{R_{m2}})}{\sqrt{\sum_n (R_{m1}[n] - \mu_{R_{m1}})^2 \sum_n (R_{m2}[n] - \mu_{R_{m2}})^2}} \quad (23)$$

where $\mu_{R_{m1}}$ and $\mu_{R_{m2}}$ are the mean values of the respective signals, k represents the lag index.

The time delay τ_{max} corresponding to the peak of the cross-correlation function provides an estimate of the phase shift between thoracic and abdominal movements

$$\tau_{\text{max}} = \arg \max_k R_{xy}[k]. \quad (24)$$

The phase difference $\Delta\phi$ at the fundamental respiratory frequency f_{resp} is then derived as

$$\varphi_{\text{corr}} = 2\pi f_{\text{resp}} \tau_{\text{max}}. \quad (25)$$

3) *FFT Method*: Alternatively, the phase difference can be characterized in the frequency domain through the following procedure: Apply Fourier transform to both displacement signals:

$$D_{m1}(f) = \mathcal{F}\{R_{m1}(\tau)\}, \quad D_{m2}(f) = \mathcal{F}\{R_{m2}(\tau)\}. \quad (26)$$

Locate the dominant respiratory frequency f_{resp} by finding the maximum magnitude in the average spectrum

$$f_{\text{resp}} = \arg \max_f |D_{m1}(f) + D_{m2}(f)|. \quad (27)$$

Extract the phase values at f_{resp} from each signal's spectrum

$$\begin{aligned} \phi_{m1}(f_{\text{resp}}) &= \arg[D_{m1}(f_{\text{resp}})], \quad \phi_{m2}(f_{\text{resp}}) \\ &= \arg[D_{m2}(f_{\text{resp}})]. \end{aligned} \quad (28)$$

TABLE I
KEY GEOMETRIC AND SIMULATION PARAMETERS

Category	Parameter	Value / Coordinate	Description
Thoracic Ellipsoid	Radii (X, Y, Z)	10, 35, 18 cm	Modeled based on average adult anthropometry.
	Center	(0, 0, 0) cm	Reference origin for the body coordinate system.
Abdominal Ellipsoid	Radii (X, Y, Z)	10, 35, 15 cm	Slightly smaller dimensions typical of the abdominal region.
	Center	(0, -20, 0) cm	Positioned to simulate a torso length (L) of 20 cm.
Radar Configuration	Position	(65, 0, 60) cm	Optimal oblique placement derived from simulation.
	Incidence Angle	$\sim 45^\circ$	Radar installation angle

Compute the phase difference between thoracic and abdominal motions

$$\varphi_{\text{FFT}} = \phi_{m2}(f_{\text{resp}}) - \phi_{m1}(f_{\text{resp}}). \quad (29)$$

In subsequent simulations and experimental analyses, we will compare the advantages and disadvantages of these three methods and finally present our conclusions.

III. SIMULATION RESULTS

Based on the established 4-D thoracoabdominal scattering model, we conducted simulations and analyses of radar echoes under different radar configuration scenarios and various thoracoabdominal breathing desynchronization conditions in MATLAB (R2023a). By observing the echo characteristics and employing the proposed demodulation and signal processing methods, we assessed the detection performance of different radar configurations in identifying thoracoabdominal breathing desynchronization across various scenarios. Furthermore, we investigated the accuracy of the proposed method in extracting thoracoabdominal breathing desynchronization phases under different conditions.

A. Detection Efficacy for Paradoxical Breathing

First, we will investigate the detection efficacy for varying degrees of asynchronous breathing when the radar wave is obliquely incident on the thoracoabdominal surface. As shown in Fig. 2, we begin by simulating a scenario where the radar wave obliquely illuminates the human thorax, and a thoracoabdominal paradoxical breathing scenario is established. In the simulation, the FMCW radar is positioned at (65, 0, 60) with a range resolution of 4 cm. The HPBW in the H -plane of the radar transmit antenna is 30° (consistent with the parameters of the subsequent practical FMCW radar prototype used in experiments), and the radar is oriented toward the coordinate origin. The key geometric and simulation parameters are summarized in Table I. Regarding the surface meshing into unit cells, as detailed in Fig. 2, each ellipsoidal surface is divided into a 30×30 grid of unit cells. The average diagonal size of all unit cells is 4.3 cm. When using an FMCW radar

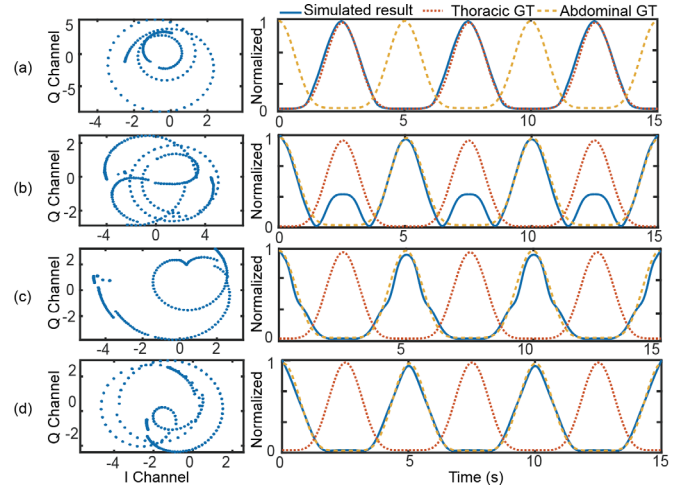


Fig. 5. Simulation results for paradoxical breathing. (a)–(d) IQ constellation diagrams (left) of the data from the 22nd, 24th, 26th, and 28th range bins, respectively, and the respiratory waveforms demodulated using the method proposed in this article (right).

operating in the 77 GHz band (wavelength $\lambda = 0.38$ cm), this mesh size satisfies the underlying assumptions for RCS estimation using the PO method.

For the dynamic parameters, the thoracic movement is defined as $x_{\text{thorax}}(t) = 1.2 \times \cos^6(\pi \times 0.2 \times t)$ cm, and the abdominal movement is defined as $x_{\text{abdomen}}(t) = -1.2 \times \cos^6(\pi \times 0.2 \times t)$ cm, simulating a paradoxical breathing pattern with a respiratory frequency of 0.2 Hz and a displacement amplitude of 1.2 cm.

Under these simulation parameter settings, the simulated echo data I-Q constellation diagrams and the demodulated respiratory waveforms from different range bins are shown in Fig. 5. All time-domain waveforms in the figure have been normalized. The relative distance between the radar and the center of the thoracic model surface is approximately 88 cm, while that for the abdominal model surface is approximately 112 cm, corresponding to the 22nd and 28th range bins of the FMCW radar, respectively. As observed in the simulation results in Fig. 5, the respiratory waveforms demodulated from these two range bins are in opposite phase, which is generally consistent with the ground truth (GT) of the predefined thoracoabdominal breathing motions. For the intermediate range bins, specifically the 24th and 26th, the trajectories in the I-Q constellation diagrams appear more complex, and the demodulated waveforms exhibit varying degrees of distortion. This phenomenon is likely attributable to the complex EM scattering resulting from the paradoxical motion in the thoracoabdominal junction region, where signals within a single range bin contain superimposed echoes from multiple moving sources.

B. Detection Efficacy for Varying Degrees of Asynchronous Breathing

To evaluate the accuracy of our proposed asynchronous phase detection method in calculating different asynchronous phases, we modified the thoracoabdominal breathing phase difference values based on the model parameters established in

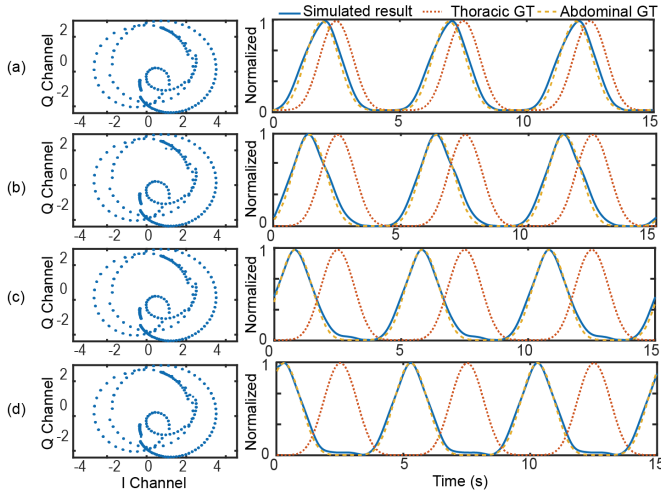


Fig. 6. Simulation results for different asynchronous thoracic and abdominal phases in the 26th range bin. (a)–(d) Represent the model settings for asynchronous thoracic and abdominal breathing phases of 0° , 40° , 80° , and 120° , respectively.

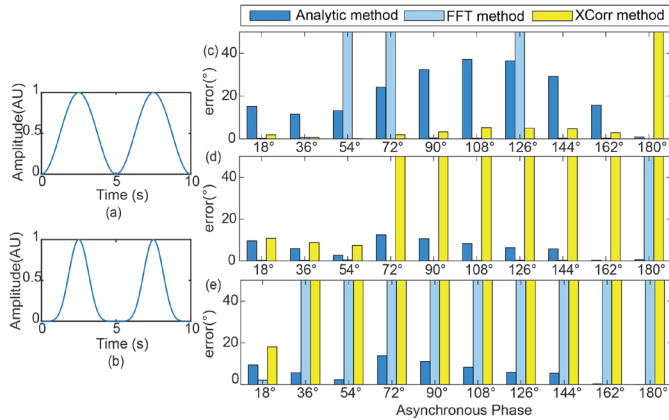


Fig. 7. Respiratory waveforms for (a) $n = 1$ and (b) $n = 3$. And the phase estimation errors for signals using analytic method, FFT method, and XCorr method with (c) 10-s time window and simulated respiratory parameter $n = 3$, (d) 10-s time window and simulated respiratory parameter $n = 1$, and (e) 4-s time window and simulated respiratory parameter $n = 1$. (For clarity, we limited the y-coordinate range to 0° – 50° ; phase values outside this range are considered to have extremely large errors.)

Section III-A. Phase estimation was performed using analytic method, FFT-based method and the cross correlation method. Fig. 6 presents the simulation results from the 26th range bin under different thoracoabdominal asynchronous phase conditions. All time-domain waveforms in the figure have been normalized. It can be observed that the demodulated abdominal respiratory waveform is largely consistent with the abdominal GT.

For a more detailed comparison of the estimation accuracy between the analytic-based, FFT-based and XCorr-based methods, we uniformly selected 10 phase values between 0° and 180° , conducting simulations and phase estimations for each. The results are shown in Fig. 7(c)–(e). In this simulation, we evaluated the asynchronous phase calculated using three different methods under varying time windows and respiratory conditions. Fig. 7(c) presents the error analysis with a 10-s time window and a respiratory waveform parameter

TABLE II
ERROR OF PHASE-DIFFERENCE ESTIMATOR WITH RESPECT TO GT IN DIFFERENT CASES

Conditions	RMSE with Analytic method	RMSE with XCorr -based method	RMSE with FFT -based method
SNR=40dB, $n=1$	7.2°	57.3°	4.7°
SNR=20dB, $n=1$	16°	62°	14.6°
SNR=10dB, $n=1$	21.3°	57.9°	57°
SNR=40dB, $n=3$	23°	5.3°	51°
SNR=20dB, $n=3$	26.4°	15.1°	56°
SNR=10dB, $n=3$	33°	62°	54°

$n = 3$. Under this respiratory parameter configuration, which features a respiratory plateau period approximating normal resting breathing characteristics, the XCorr method demonstrated the highest accuracy. The FFT method exhibited a specific outlier deviation, while the analytic method showed relatively large overall errors. This is likely because the analytic method was originally derived under the respiratory parameter $n = 1$, and when applied to $n = 3$ conditions, it no longer satisfies the analytic computation requirements, resulting in increased errors.

Fig. 7(d) shows the error analysis results with a 10-s time window and respiratory waveform parameter $n = 1$. Here, the XCorr method displayed significant errors, confirming its unsuitability for phase difference calculation. Meanwhile, the analytic method's error decreased substantially, consistent with our initial hypothesis, while the FFT method maintained minimal error.

Since real-world scenarios may involve time windows that aren't integer multiples of respiratory cycles, and since apnea events can be brief, we simulated shorter, noninteger multiple time windows to validate the robustness of these three methods. Fig. 7(e) displays the error analysis with a 4-s time window and respiratory parameter $n = 1$, where the FFT error increased sharply while the analytic method's error remained relatively stable. This occurs because the FFT method fundamentally relies on accurate frequency estimation, and its performance is heavily influenced by time window duration, thus lacking strong robustness.

Moreover, we applied additive white Gaussian noise (AWGN) with SNR levels of 40, 20, and 10 dB to the simulated 10-s I/Q baseband signals. The accuracy of each algorithm was evaluated by calculating the root mean square error (RMSE) of the estimated phase difference across a full range of asynchronous phases (0° – 180° with a 10° interval). The quantitative results for different respiratory morphologies and SNR conditions are presented in Table II.

Synthesizing these results with the analysis previously presented in Section III, we have summarized the pros and cons of the three methods in Table III. This comparison highlights the specific advantages of the proposed analytic method in

TABLE III
COMPARISON OF PHASE ESTIMATION METHODS

Method	Analytic method	FFT-based method	XCorr-based method
Waveform morphology robustness	Moderate: Stable performance for clinically common short plateau waveforms ($n=1$).	Moderate: Limited by spectral leakage, it has weak ability to handle peak waveforms with non-integer periods.	Poor: Heavily dependent on obvious time-domain characteristics, it fails under symmetrical waveforms with $n=1$.
Low SNR/short window robustness	Excellent: Can still track phase changes under short time windows and noisy environments.	Poor resolution under short time windows (e.g., 4s), the error increases exponentially.	Poor: Significantly affected by noise, peak alignment is easily interfered with.
Core application scenarios	Practical phase monitoring in real-world environments.	Long-term, steady-state respiratory monitoring in ideal environments.	Simple waveform delay estimation, not suitable for complex respiratory patterns.

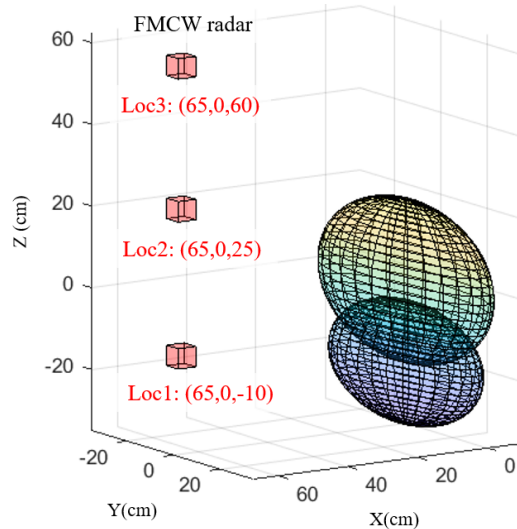


Fig. 8. Simulation of different radar locations on detection of thoracoabdominal paradoxical breathing.

handling nonsinusoidal waveforms and lower SNR conditions typical of real-world environments.

C. Impact of Radar Placement on the Detection of Thoracoabdominal Paradoxical Breathing

We further investigate the impact of different radar positions on the detection of thoracoabdominal paradoxical breathing. As shown in Fig. 8, three different radar placements were simulated using the established 4-D thoracoabdominal scattering model. For all positions, the radar was oriented toward the new coordinate origin. All time-domain waveforms in the figure have been normalized. Position Loc1 represents the radar facing the thoracoabdominal region approximately head-on, which is the typical configuration adopted in most previous FMCW radar-based thoracoabdominal respiration monitoring

studies [20]. Positions Loc2 and Loc3, in contrast, correspond to oblique illumination of the thoracoabdominal region.

Fig. 9 presents the 15-s simulated radar echoes for these three radar positions. The upper subplots all show the equivalent baseband signal results from the range bin corresponding to the chest, while the lower subplots show the equivalent baseband signal results from the range bin corresponding to the abdomen. In addition, their raw I/Q constellation diagrams along with the demodulated respiratory waveforms are displayed. Fig. 9(a) shows the results for position Loc1, specifically data from the 11th and 14th range bins. It can be observed that the raw I-Q diagrams for these range bins are relatively complex, making it difficult to discern the antiphase motion, and the demodulation error is significant. Fig. 9(b) and (c) corresponds to the results for positions Loc2 and Loc3, respectively. For Loc3, the data are from the 22nd and 28th range bins, while for Loc2, they are from the 19th and 26th range bins. In both of these oblique incidence cases, the characteristic patterns of paradoxical breathing are observable.

We also conducted an analysis of the estimation error for asynchronous breathing phases ranging from 1° to 180° under these three radar placement scenarios using the FFT-based method. The results are shown in Fig. 10. Since direct frontal illumination of the thoracoabdominal region makes it difficult to distribute the thoracic and abdominal components into separate range bins, it introduces more demodulation and estimation errors. In contrast, oblique incidence increases the radial separation between the thoracic and abdominal surfaces, reducing the number of overlapping motion components within each range bin. Consequently, the configuration with the largest elevation angle (Loc3) yields the smallest asynchronous-phase estimation error.

D. Discussion on the Impact of Body Shape Variations

The fundamental prerequisite for decoupling respiratory signals is that the scattering centers of the thorax and abdomen must be resolved into distinct range bins. This requires their radial distance difference ΔR along the radar's line-of-sight (LOS) to exceed the radar's range resolution d_{res} .

For the general experimental setup as shown in Fig. 11(a), based on geometric relationships, the radial distance difference ΔR between the chest and abdomen relative to the radar can be simplified as

$$\Delta R \approx L \cdot \sin \theta. \quad (30)$$

For a typical adult ($L \approx 20 - 30$ cm), at $\theta = 45^\circ$, ΔR ranges from 14.1 to 21.2 cm, which is 3.4–5.1 times the d_{res} [27]. Even for a smaller subject ($L = 15$ cm), $\Delta R \approx 10.6$ cm $\approx 2.5d_{\text{res}}$. This indicates a significant tolerance margin for varying body types.

The $\sin \theta$ function is relatively stable around 45° . A $\pm 10^\circ$ installation error (at $\theta = 35^\circ$ or 55°) still maintains ΔR well above d_{res} .

Variations in ellipsoid sizes primarily affect the RCS magnitude and the specific range bin index but do not inherently prevent separation.

IV. EXPERIMENTAL VALIDATION

To practically validate the simulation analysis, we conducted human experiments to demonstrate the practical feasibility

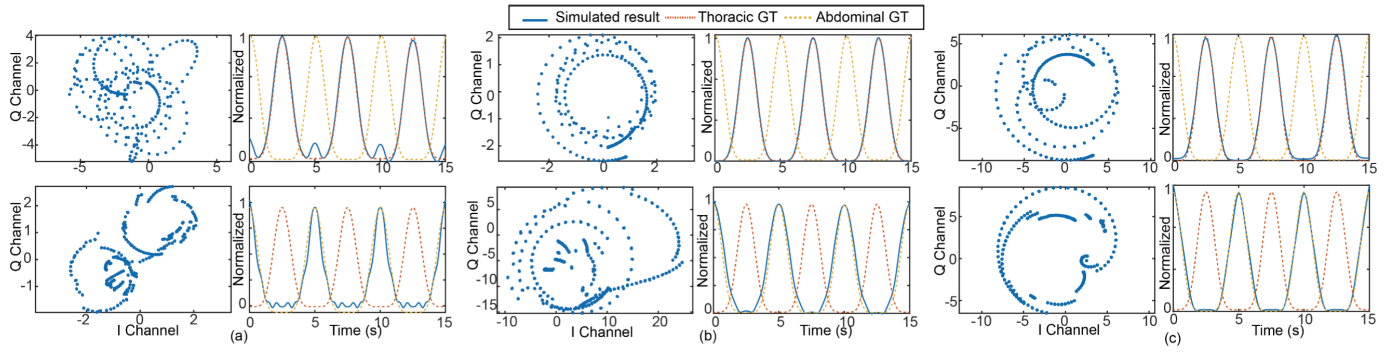


Fig. 9. Simulated results for three different radar positions shown in Fig. 8. (a) For Loc1, (b) for Loc2, and (c) for Loc3.

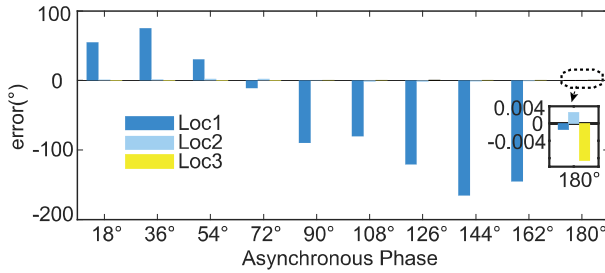


Fig. 10. Estimation error versus asynchronous phase for Loc1/Loc2/Loc3.

TABLE IV
RADAR OPERATIONAL PARAMETERS

Parameter	Value
Start Frequency	77.0 GHz
Effective Bandwidth	3646 MHz
Frequency Slope	28.49 MHz/ μ s
Range Resolution	4.11 cm
ADC Samples per Chirp	256
ADC Sampling Rate	2 Msps
Chirp Duration	140.26 μ s
Idle Time	7 μ s
Chirps per Frame	8
Frame Periodicity	10 ms (100 Hz)
Antennas used in this system	1 TX / 1 RX

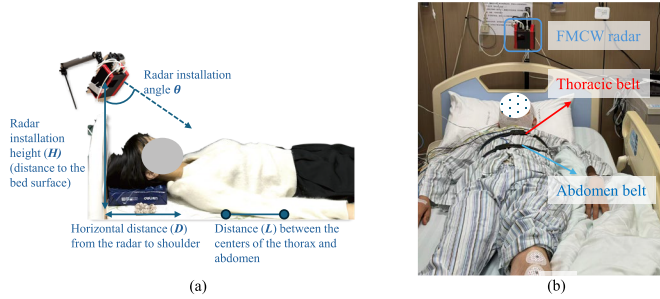


Fig. 11. (a) Schematic illustrating the oblique-incidence geometry of the radar sensor relative to the supine participant, with key geometric parameters labeled. (b) Photograph of the radar system mounted on the headboard of a hospital bed during a simultaneous, in-laboratory PSG study.

of the proposed method for detecting respiratory asynchrony phase. Since asynchronous breathing typically does not occur spontaneously in healthy individuals during daytime activities and is difficult to voluntarily mimic, we attempted to detect abnormal thoracic and abdominal breathing phenomena by collecting data throughout the night from individuals who were highly likely to experience OSA.

A. Hardware Description and Preprocessing

The noncontact sensing apparatus consisted of a commercial-off-the-shelf (COTS) 77 GHz radar system. The core of the system is a Texas Instruments AWR2243BOOST evaluation module, a single-chip FMCW sensor operating in the 77-GHz band [25]. The module features three TX and four RX antennas; however, we only used one TX and one RX mode for experimental validation.

The system utilizes the onboard antennas of the AWR2243BOOST evaluation board. These are linearly polarized, 3-element series-fed patch antennas. According to the datasheet [25], the antenna features a HPBW of approximately 30° in the H plane. In our experimental configuration, the antenna's H plane was aligned with the longitudinal axis of the human body. This orientation ensures that the beamwidth is sufficient to cover the entire thoracoabdominal region of the subject. The specific geometric parameters are detailed in Fig. 11(a). The radar was mounted at a vertical height of 65 cm above the mattress. Due to the oblique incidence geometry ($\theta = 45^\circ$), the LOS distance was approximately 80 cm to the thoracic center and 100 cm to the abdominal center.

Raw ADC data from the sensor was captured using a Texas Instruments DCA1000EVM data capture card. The DCA1000EVM streams the high-speed LVDS data from the AWR2243 to a host personal computer (PC) via an Ethernet connection for offline storage and signal processing. All subsequent analysis, including the implementation of the demodulation and scattering models described herein, was performed in MATLAB (R2023a). The specific operational parameters of the FMCW radar were configured for the clinical study as detailed in Table IV.

In terms of signal preprocessing, since we are using a 1T1R FMCW radar, calibration between different channels is not required. However, to enhance the SNR of the baseband signal, we selected the middle four chirp signals from the eight chirps within the same frame of the FMCW radar during preprocessing and averaged them to improve the SNR.

For the selection of range bins corresponding to the chest and abdomen, since the experimental setup is fixed, the theoretical LOS distances to the subject's chest and abdomen can be predetermined. This allows us to define specific regions of interest in the range profile where the target signals are expected to appear. Subsequently, we further analyzed the range-FFT profile of the radar echo. The chest and abdomen manifest as distinct local maxima in the range-FFT profile, after which we manually selected the range bins for the chest and abdomen. The current focus of the work lies in preliminarily validating the estimation of the chest-abdominal phase difference, and an automatic rangebin-selection functionality has not yet been integrated into the signal processing workflow.

B. Model Validation

To quantify the consistency between simulation and experimental data, we collected real-world measurements from a healthy female subject in a laboratory setting and used the corresponding physical parameters to drive our simulation.

The specific experimental scenario is illustrated in Fig. 11(a). The radar was mounted at a height (H) of 65 cm with an oblique incidence angle θ of 45° . The horizontal distance (D) from the radar to the subject's shoulder was 60 cm, and the distance (L) between the centers of the thorax and abdomen was approximately 20 cm. During the experiment, the subject maintained normal synchronous breathing. For the simulation, model parameters were strictly aligned with these physical settings. Specifically, the respiratory motion equations for the thorax and abdomen were derived by inverting the actual displacement data demodulated from the experimental range bins.

The comparison between experimental and simulated results is presented in Fig. 12. The left panels display the raw I/Q signals from the respective thoracic and abdominal range bins, while the right panels show the corresponding range-time profiles calculated from the complex I/Q data. Thoracic Results: Fig. 12(a) and (b) shows the experimental raw I/Q signal and range-time profile, respectively, while Fig. 12(c) and (d) presents the simulated counterparts. Abdominal Results: Fig. 12(e) and (f) depicts the experimental raw I/Q signal and range-time profile, while Fig. 12(g) and (h) shows the simulated versions.

Visually, the morphology of the range-time profiles from the experiment and simulation shows a high degree of similarity, confirming that the 4-D model accurately captures the scattering dynamics.

To quantitatively evaluate the similarity between the raw I/Q signals, we calculated the Pearson's correlation coefficients between the experimental and simulated waveforms: Thorax: The correlation coefficients for the I and Q channels were 0.74 and 0.78, respectively. Abdomen: The correlation coefficients for the I and Q channels were 0.71 and 0.73, respectively.

Considering that real-world human respiration is not an ideal motion characterized by constant amplitude and periodicity, and that the specific anatomical features of the female subject do not perfectly align with the simplified ellipsoid model, these correlation results (exceeding 0.7) serve to validate the accuracy of the proposed model to a certain extent.

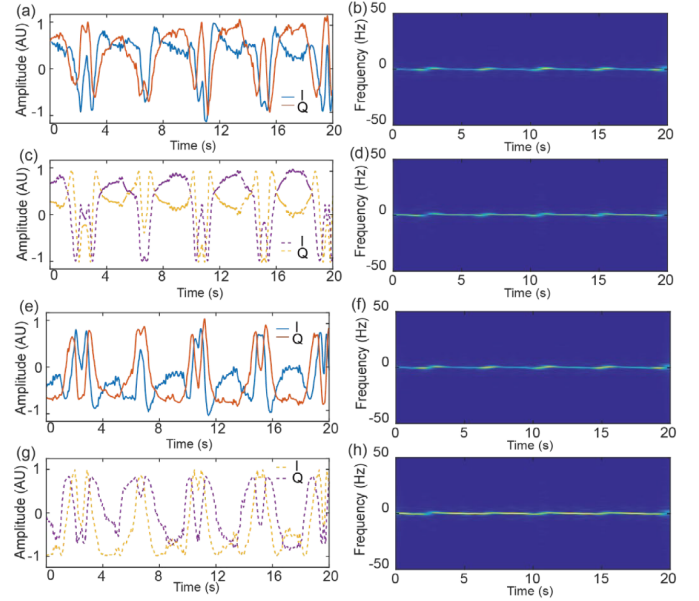


Fig. 12. Comparative analysis of experimental and simulated radar echoes. (a) and (c) Raw I/Q signals for the thoracic range bin from experimental data and simulation, respectively. (b) and (d) Corresponding range-time profiles for the thoracic region, illustrating the displacement fidelity of the model. (e) and (g) Raw I/Q signals for the abdominal range bin from experimental data and simulation, respectively. (f) and (h) Corresponding range-time profiles for the abdominal region.

C. Sleep Study Protocol and Data Acquisition

The sleep study was conducted in collaboration with the Sleep Department of the First Affiliated Hospital of Soochow University. The study protocol received approval from the hospital's Institutional Review Board, and written informed consent forms were obtained from the participants. Ten patients (seven male, three female; mean age: 52.3 ± 13.4 years; mean BMI: 24.1 ± 3.2 kg/m²) with varying degrees of OSA severity were enrolled in this study. The cohort included one patient without OSA ($AHI < 5$), three with mild OSA ($5 \leq AHI < 15$), two with moderate OSA ($15 \leq AHI < 30$), and four with severe OSA ($AHI \geq 30$). Considering the unavoidable effects of body movement, we removed interfering time periods such as body movement during data processing. Subject characteristics are summarized in Table V.

During the sleep experiment, the participant underwent a full, attended, in-laboratory overnight PSG, which served as the clinical gold standard. Physiological signals were acquired using a Compu medics Graef HD-PSG system, a 24-bit resolution amplifier for AASM Type 1 sleep studies. Two dedicated RIP belts placed around the participant's thorax and abdomen. Nasal airflow pressure and nasal thermal sensors are also placed according to operating standards, serving as the gold standard for diagnosing sleep apnea [26]. Following data acquisition, all PSG recordings were scored by hospital physicians for sleep stages and respiratory events according to AASM guidelines.

Concurrently, the FMCW radar system was deployed at the patient's bedside, as illustrated in Fig. 11. The radar sensor was affixed to an articulated arm clamped to the headboard of

TABLE V
PARTICIPANT DEMOGRAPHICS AND PHASE DETECTION PERFORMANCE

Subject	Gender	Age (year)	BMI	AHI (events/h)	OSA Severity	Epochs Analyzed	Correlation
1	M	52	26.8	36.3	Severe	876	0.80
2	F	40	21.5	8.5	Mild	842	0.89
3	M	60	25.2	53.0	Severe	756	0.77
4	M	77	21.0	14.2	Mild	912	0.81
5	M	50	25.1	19.7	Moderate	854	0.83
6	M	53	28.6	75.1	Severe	798	0.75
7	F	30	19.7	0.7	None	894	0.92
8	M	40	28.1	68.4	Severe	948	0.76
9	F	61	20.5	12.9	Mild	826	0.84
10	M	60	24.2	16.0	Moderate	762	0.78
Average	—	52.3	24.1	30.5	—	846.8	0.82±0.06

the hospital bed. It was positioned at a height of approximately 65 cm above the mattress and centered relative to the patient's torso, with a downward-facing angle of 45°. This oblique incidence geometry is fundamental to the proposed method, establishing a LOS distance of approximately 80 cm to the thorax and 100 cm to the abdomen. This geometric separation ensures that the chest and abdomen are resolved into distinct range bins, allowing for the noncontact detection of their independent movement. Raw radar data and clinical PSG data were recorded continuously throughout the entire sleep period. Temporal synchronization between the two data streams was achieved by ensuring the clocks of the PSG system and the radar's host PC were synchronized prior to each recording. The resulting timestamped datasets were then precisely aligned in post-processing.

D. Asynchronous Thoracoabdominal Breathing Detection

We performed a detailed analysis of overnight data from a participant who experienced multiple episodes of OSA. The radar data, synchronously collected with PSG, underwent preprocessing where the approximate range bin region (22)–(25) corresponding to thoracic and abdominal movements was identified based on prior knowledge. The proposed thoracoabdominal motion demodulation algorithm was then applied to extract raw waveforms from range bins 22–25. Fig. 13 displays 30 s of synchronized radar data and PSG recordings. Fig. 13(a)–(d) shows the IQ constellation diagrams (left) of raw data from different range bin channels alongside their corresponding demodulated motion waveforms (right). Fig. 13(f) presents the thermal sensor signal acquired by PSG during the same period, containing an obstructive apnea event annotated by sleep specialists. All time-domain waveforms in the figure have been normalized.

By comparing the measured radar signal's IQ constellation diagrams with the simulation results in Fig. 5, it can be observed that the actual measurements from the thoracic and abdominal regions closely resemble the IQ diagram patterns [left side of Fig. 5(a) and (d)] from the simulation where thorax and abdomen existed separately in their respective range bins. However, due to parameter discrepancies between

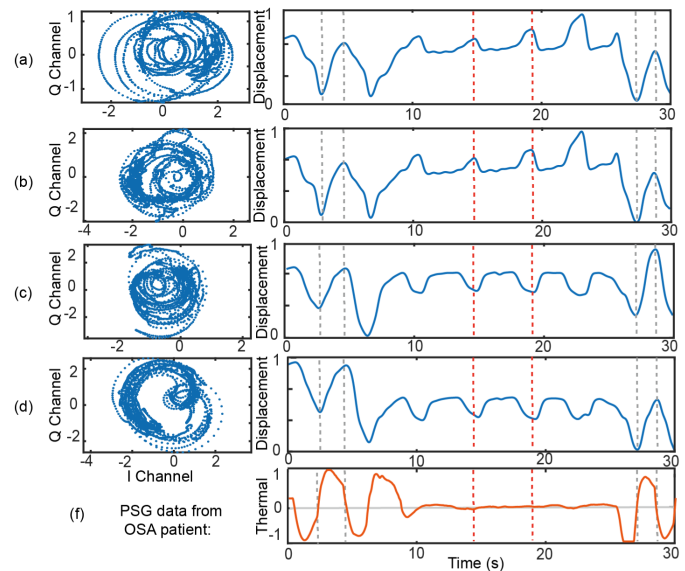


Fig. 13. Experimental results of different range bins. (a)–(d) Display the IQ constellation diagrams (left) of raw data from 22 to 25 range bin channels and their corresponding demodulated motion waveforms (right), respectively. (f) Presents the thermal sensor signal from PSG during the same interval, containing an obstructive apnea event annotated by sleep specialists.

the simulation and actual patient anatomy, the IQ constellation diagrams from the two middle range bins don't show similar consistency.

Analysis of the radar waveforms in Fig. 13 reveals that the demodulated motion waveforms from range bins 22 and 23 (thoracic region) maintain consistent morphology and phase, while those from bins 24 and 25 (abdominal region) demonstrate similar morphological consistency. However, these two signal groups exhibit clear phase inversion during the apnea episode.

When examining normal breathing periods, the respiratory peaks marked by gray vertical dashed lines in Fig. 13 align precisely with the zero-crossing points in the thermal sensor signal-representing the momentary pause in airflow at the end of inhalation and exhalation. This correspondence validates the accuracy of the radar detection method.

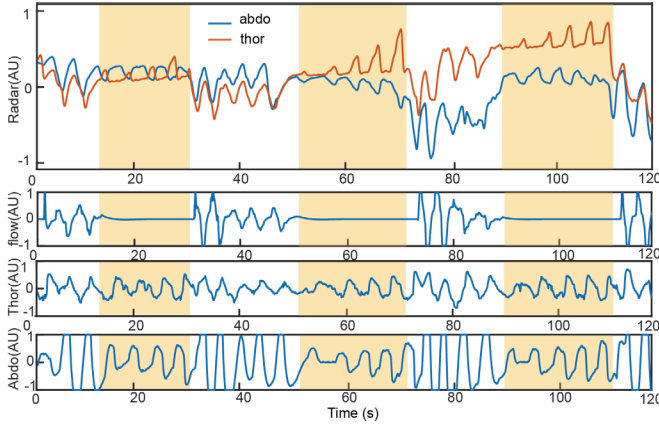


Fig. 14. 2-min synchronized data comparison between PSG and radar signals. The yellow shaded regions designate three episodes of obstructive apnea, as identified by the sleep technologist.

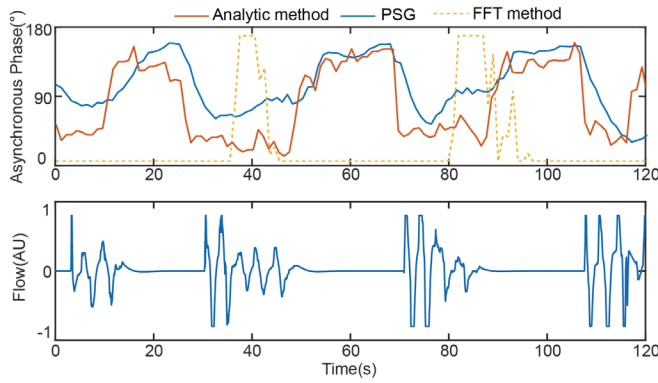


Fig. 15. Asynchronous phase calculation results using analytic method and FFT method, compared with PSG signals.

Fig. 14 presents a 2-min synchronized data comparison between PSG and radar measurements from the OSA patient. The highlighted regions indicate three OSA events annotated by sleep technicians. During these OSA episodes, the PSG signals reveal clear phase opposition between thoracic and abdominal belt signals, accompanied by complete cessation of nasal airflow. Correspondingly, the radar-acquired thoracic and abdominal signals demonstrate distinct phase inversion during these three apnea events, while maintaining phase synchronization during normal breathing periods. This strong agreement with gold-standard signals validates the effectiveness of the proposed methodology.

E. Asynchronous Phases Analysis

We performed thoracoabdominal asynchronous phase analysis and calculation on the data collected during the aforementioned period, using raw data from PSG thoracic and abdominal belts as the gold standard for comparison. Fig. 15 presents the continuous phase detection results, where the gold-standard phase difference was obtained by applying the same phase calculation method to the PSG belt data.

The results demonstrate that the analytic method shows the highest agreement with the gold standard in actual human experiments. Nevertheless, certain errors persist, partly

attributable to the susceptibility of chest belt data to motion artifacts from limb movements. Additionally, as observed in the abdominal channel data in the bottom panel of Fig. 13, signal clipping distortion occurs during certain periods, introducing inherent errors even in the gold-standard calculations.

Despite these minor discrepancies, comparison with the respiratory flow data (gold standard) in Fig. 14 confirms that radar-calculated phase asynchrony remains low (approaching 0°) during normal breathing, while increasing significantly (approaching 180°) during OSA events. This pattern accurately reflects the characteristic paradoxical breathing associated with OSA.

To quantify the agreement between radar-derived and PSG-derived phase differences, the Pearson's correlation coefficient (r) was computed for each subject across nonoverlapping 30-s epochs. Let $\varphi_{\text{radar}}^{(i)}$ and $\varphi_{\text{PSG}}^{(i)}$ denote the thoracoabdominal asynchronous phase estimated from radar and PSG signals for epoch i , respectively. The correlation coefficient was computed as

$$r = \frac{\sum_{i=1}^N (\varphi_{\text{radar}}^{(i)} - \overline{\varphi_{\text{radar}}}) (\varphi_{\text{PSG}}^{(i)} - \overline{\varphi_{\text{PSG}}})}{\sqrt{\sum_{i=1}^N (\varphi_{\text{radar}}^{(i)} - \overline{\varphi_{\text{radar}}})^2} \cdot \sqrt{\sum_{i=1}^N (\varphi_{\text{PSG}}^{(i)} - \overline{\varphi_{\text{PSG}}})^2}} \quad (31)$$

where $\overline{\varphi_{\text{radar}}}$ is average of radar-derived phase difference, $\overline{\varphi_{\text{PSG}}}$ is average of PSG-derived phase difference and N is the total number of epochs analyzed per subject.

A total of 8468 epochs (mean: 846.8 ± 63.2 per subject) were analyzed. As shown in Table V, the proposed method achieved a mean correlation of $r = 0.82 \pm 0.06$. Subjects with clearer respiratory patterns exhibited higher correlations ($r > 0.84$), while those with severe OSA showed lower correlations, consistent with increased signal complexity during severe apnea episodes.

The lower correlation in severe OSA cases primarily reflects the physiological complexity of the event rather than a fundamental failure of the radar sensor itself. Severe OSA episodes are frequently accompanied by irregular, nonsinusoidal breathing and, more critically, involuntary body movements (arousals or struggling). These body movements introduce artifacts that affect both the radar signal and the reference PSG respiratory belts (which are susceptible to displacement). The current medical gold standard for determining OSA events still relies on synchronized data from multiple sensors, including nasal airflow, nasal thermistors, and respiratory belts. This is because belts may slip or provide ambiguous phase data during respiratory difficulty, whereas the zero-crossing points (indicating airflow cessation/initiation) from thermistor and nasal airflow sensors provide precise temporal markers for the onset and termination of apnea events.

V. CONCLUSION AND DISCUSSION

Building on the mechanism of thoraco-abdominal breathing, this article proposes a novel 4-D thoracic backscattering model that approximates independent chest and abdominal motions. Tailored to FMCW sensing, we develop a theoretical simulation framework for echoes under oblique thoracic illumination, validating the feasibility of using multirange-bin channels to demodulate separate respiratory movements.

The model provides realistic simulated data to support radar-based respiratory monitoring, particularly for detecting abnormal breathing patterns. Below, we discuss limitations and future directions.

To improve radar detection accuracy during severe OSA events, a threshold can be set during the preprocessing stage to identify and exclude data segments containing significant body movement, thereby avoiding motion artifacts interfering with phase relationships. Regarding algorithm selection, analytical methods outperform FFT-based methods in handling typical short-duration nonsinusoidal waveforms in severe OSA events. Future research could further improve the accuracy of phase detection algorithms for specific breathing waveform patterns in OSA. Additionally, an installation angle of 45° is recommended to balance spatial separation and signal strength, as this optimal angle maximizes the separation of chest and abdominal signals, ensuring the signal-to-noise ratio of independent channel demodulation.

The proposed model relies on a simplified geometry of an average adult and was validated with supine subjects under controlled conditions. It does not account for anatomical diversity, gross body motion, or posture changes. Future work should improve anatomical fidelity and develop motion-artifact compensation algorithms for real-world use.

Clinical validation remains preliminary, based on case-level evidence from a limited cohort. Although radar-derived phase differences showed epochwise correlation with PSG signals, diagnostic accuracy across a larger, diverse population is not yet established. A large-scale clinical trial is needed to confirm diagnostic utility.

Performance depends on radar-subject geometry. This study tested only three discrete radar placements; full angular optimization was beyond its scope. Future studies should systematically explore incidence angles and distances to define robust operating regimes for varied clinical scenarios.

ACKNOWLEDGMENT

The authors thank the sleep technicians from the Sleep Department of The First Affiliated Hospital of Soochow University for their help and guidance during the human trials, as well as for their data annotation work.

REFERENCES

- [1] J. Hammer and C. J. L. Newth, "Assessment of thoraco-abdominal asynchrony," *Paediatric Respiratory Rev.*, vol. 10, no. 2, pp. 75–80, Jun. 2009.
- [2] P.-H. Huang, W.-C. Chung, C.-C. Sheu, J.-R. Tsai, and T.-C. Hsiao, "Is the asynchronous phase of thoracoabdominal movement a novel feature of successful extubation? A preliminary result," in *Proc. 43rd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Nov. 2021, pp. 752–756.
- [3] C. P. Cri e et al., "Body plethysmography—Its principles and clinical use," *Respiratory Med.*, vol. 105, no. 7, pp. 959–971, Jul. 2011.
- [4] N. L. Loo, Y. S. Chiew, C. P. Tan, G. Arunachalam, A. M. Ralib, and M.-B. Mat-Nor, "A machine learning model for real-time asynchronous breathing monitoring," *IFAC-PapersOnLine*, vol. 51, no. 27, pp. 378–383, 2018.
- [5] L. Wen et al., "Noncontact infant apnea detection for hypoxia prevention with a K-band biomedical radar," *IEEE Trans. Biomed. Eng.*, vol. 71, no. 3, pp. 1022–1032, Mar. 2024.
- [6] M. Baboli, A. Singh, B. Soll, O. Boric-Lubecke, and V. M. Lubecke, "Wireless sleep apnea detection using continuous wave quadrature Doppler radar," *IEEE Sensors J.*, vol. 20, no. 1, pp. 538–545, Jan. 2020.
- [7] J. E. Kiriazi, O. Boric-Lubecke, and V. M. Lubecke, "Modeling of human torso time-space characteristics for respiratory effective RCS measurements with Doppler radar," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Baltimore, MD, USA, Jun. 2011, pp. 1–4.
- [8] J. N. Sameera, A. Lee, V. M. Lubecke, and O. Boric-Lubecke, "Respiratory dynamics of thoracic and abdominal motion in Doppler radar measurements," in *Proc. IEEE Radio Wireless Symp. (RWS)*, Jan. 2024, pp. 39–41.
- [9] S. Dong et al., "Doppler cardiogram: A remote detection of human heart activities," *IEEE Trans. Microw. Theory Techn.*, vol. 68, no. 3, pp. 1132–1141, Mar. 2020.
- [10] N. Jiang et al., "Separation of thoracic and abdominal measurements in neonatal vital sign monitoring using radar vision fusion system," *Sci. Rep.*, vol. 15, no. 1, p. 11800, Apr. 2025.
- [11] F. Zhang, Z. Zhang, L. Kang, A. Zhou, and H. Ma, "MmTAA: A contact-less thoracoabdominal asynchrony measurement system based on mmWave sensing," *IEEE Trans. Mobile Comput.*, vol. 24, no. 2, pp. 627–641, Feb. 2025.
- [12] S. Kang et al., "Non-contact diagnosis of obstructive sleep apnea using impulse-radio ultra-wideband radar," *Sci. Rep.*, vol. 10, no. 1, p. 5261, Mar. 2020.
- [13] J. W. Choi et al., "Automated detection of sleep apnea-hypopnea events based on 60 GHz frequency-modulated continuous-wave radar using convolutional recurrent neural networks: A preliminary report of a prospective cohort study," *Sensors*, vol. 22, no. 19, p. 7177, Sep. 2022.
- [14] C. Li, S. Zheng, X. Hui, and X. Zhang, "Electromagnetic effective coupling and high spatial resolution theory in near-field cardiac sensing," *IEEE Trans. Microw. Theory Techn.*, vol. 72, no. 4, pp. 2283–2297, Apr. 2024.
- [15] G. Xue, J. Liu, and C. Gu, "Kinetic human torso ray tracing model for vital signs extraction," in *IEEE MTT-S Int. Microw. Symp. Dig.*, May 2025, pp. 1–3.
- [16] Y. Lu, C. Gu, L.-S. Wu, and J.-F. Mao, "A 3-D ray tracing model for short-range radar sensing of hand gestures," in *Proc. IEEE Asia-Pacific Microw. Conf. (APMC)*, Hong Kong, Dec. 2020, pp. 1107–1109.
- [17] K. Ashutosh, R. Gilbert, J. H. Auchincloss, and D. Peppi, "Asynchronous breathing movements in patients with chronic obstructive pulmonary disease," *Chest*, vol. 67, no. 5, pp. 553–557, May 1975.
- [18] J. Kohyama, T. Shiiki, M. Shimohira, and T. Mimura, "Asynchronous breathing during sleep," *Arch. Disease Childhood*, vol. 84, no. 2, pp. 174–177, 2001.
- [19] S. Dong, Y. Li, J. Lu, Z. Zhang, C. Gu, and J. Mao, "Accurate detection of Doppler cardiograms with a parameterized respiratory filter technique using a K-band radar sensor," *IEEE Trans. Microw. Theory Techn.*, vol. 71, no. 1, pp. 71–82, Jan. 2023.
- [20] V. Chen, *The Micro-Doppler Effect in Radar*, 2nd ed. Norwood, MA, USA: Artech House, 2019.
- [21] J. E. Kiriazi, O. Boric-Lubecke, and V. M. Lubecke, "Radar cross section of human cardiopulmonary activity for recumbent subject," in *Proc. IEEE Annu. Int. Conf. Eng. Med. Biol. Soc. Conf.*, Sep. 2009, pp. 4808–4811.
- [22] D. Andreuccetti, R. Fossi, and C. Petrucci, "An Internet resource for the calculation of the dielectric properties of body tissues in the frequency range 10 Hz–100 GHz," IFAC-CNR, Florence, Italy, 1997. [Online]. Available: <http://niremf.ifac.cnr.it/tissprop/>
- [23] D. K. Barton, *Radar Equations for Modern Radar*. Norwood, MA, USA: Artech House, 2013.
- [24] C. Zhou et al., "Reconstructing asynchrony for mechanical ventilation using a hysteresis loop virtual patient model," *Biomed. Eng. OnLine*, vol. 21, no. 1, p. 16, Mar. 2022.
- [25] Texas Instruments. *AWR2243BOOST Evaluation Module*. Texas Instruments Website. [Online]. Available: <https://www.ti.com/tool/AWR2243BOOST>
- [26] A. Sabil et al., "Comparison of apnea detection using oronasal thermal airflow sensor, nasal pressure transducer, respiratory inductance plethysmography and tracheal sound sensor," *J. Clin. Sleep Med.*, vol. 15, no. 2, pp. 285–292, Feb. 2019.
- [27] R. O. Moody, "The position of the abdominal viscera in healthy, young British and American adults," *J. Anatomy*, vol. 61, no. 2, p. 223, 1927.